

Design Proposal: Performance of Mach-Zehnder Interferometer with different path length

Ka My Dang, Max Planck Institute of Microstructure Physics, Germany

Abstract: This design proposal investigates the performance of Mach–Zehnder interferometers (MZIs) with different path-length differences, $\Delta L = 70 \mu\text{m}$, $170 \mu\text{m}$, and $470 \mu\text{m}$, to characterise the relationship between path-length difference and free spectral range (FSR) for optical modulation applications. The MZI structures are based on strip waveguides and are designed to operate around a wavelength of 1550 nm . Simulation results are presented and compared for the three MZI configurations.

I. Introduction

The Mach–Zehnder interferometer (MZI) is a photonic device used to measure or control the relative phase difference between two optical signals originating from the same source. In the proposed design, several MZI structures are investigated. Each structure consists of an input grating coupler connected to a Y-branch, which splits the incoming light into two waveguide arms. After propagating through the two arms, the optical signals are recombined using a second Y-branch and directed to an output grating coupler. Any difference in optical phase accumulated between the two arms results in interference at the output, leading to a change in the transmitted optical power. The simulated optical properties of the strip waveguide are then used as input parameters in Lumerical INTERCONNECT to model and simulate the performance of the complete MZI circuit.

II. Theory of the Mach-Zehnder Interferometer

MZI operates by splitting an optical signal into two paths, allowing the signals to accumulate different phases, and then recombining them to produce interference.

Consider an input optical signal with intensity I_i and electric-field amplitude E_i . An ideal 50:50 Y-branch splitter divides the optical power equally between the two arms of the MZI. Therefore,

$$I_1 = I_2 = \frac{I_i}{2}$$

and because optical intensity is proportional to the square of the electric-field amplitude,

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}$$

As the light propagates through the two waveguide arms, it accumulates phase. For a lossless waveguide, the fields arriving at the output Y-branch can be written as

$$E_{01} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} ,$$

$$E_{02} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} ,$$

where L_1 and L_2 are the lengths of the two MZI arms and β is the propagation constant,

$$\beta = \frac{2\pi n_{eff}}{\lambda} .$$

Here, n_{eff} is the effective refractive index of the waveguide mode and λ is the operating wavelength.

At the second Y-branch, the two optical fields are recombined. Assuming an ideal 50:50 combiner, the output field is

$$E_0 = \frac{E_{01} + E_{02}}{\sqrt{2}} = \frac{E_i}{\sqrt{2}} (e^{-i\beta L_1} + e^{-i\beta L_2}) .$$

The corresponding output intensity is therefore

$$I_o = \frac{I_i}{4} |e^{-i\beta L_1} + e^{-i\beta L_2}|^2$$

which can be simplified to: $I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)]$

where the relative phase difference between the two arms is: $\Delta\phi = \beta \Delta L$

This equation describes the basic operating principle of the MZI. When the two signals arrive in phase ($\Delta\phi = 2m\pi$), constructive interference occurs and the output power is maximum. When they arrive out of phase ($\Delta\phi = (2m + 1)\pi$), destructive interference occurs and the output power is minimum.

For the proposed MZI designs, changing the path-length difference ΔL changes how rapidly the phase difference varies with wavelength. Consequently, it changes the spacing between adjacent transmission maxima or minima, known as the free spectral range (FSR). For a waveguide MZI, the FSR around a centre wavelength λ_0 can be approximated as

$$FSR = \frac{\lambda_0^2}{n_g \Delta L}$$

where n_g is the group index of the strip waveguide. Thus, a larger ΔL produces a smaller FSR. This relationship can be investigated using the proposed values of 70 μm , 170 μm , and 470 μm .

III. Modeling and Simulation

Waveguide

In the design, we use strip waveguide with 0.5 μm width and 0.22 μm height. The simulated waveguide mode profile is presented in the following figures:

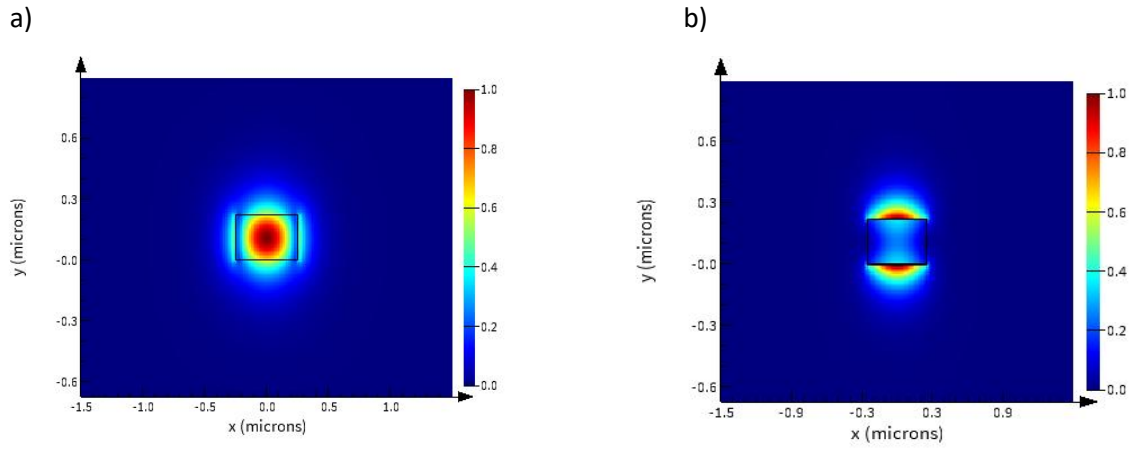


Figure 1: Electric field intensity of waveguide in TE mode (a) and TM mode (b)

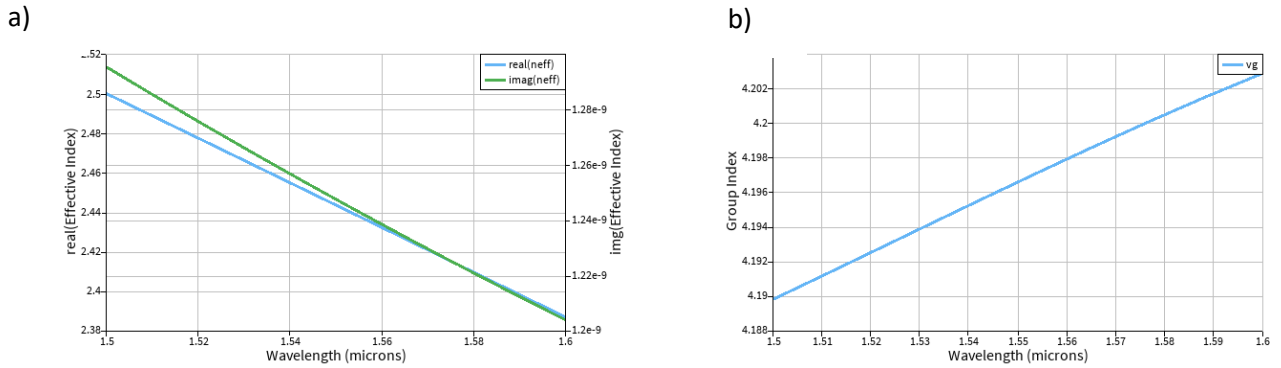


Figure 2: a) Effective index and b) group index of waveguide versus wavelength

Mach-Zehnder interferometer

The free spectra range (FSR) of the interferometer can be written:

$$FSR = \frac{\lambda_0^2}{n_g \Delta L}$$

The FSR corresponds to different ΔL in the design is shown in Table 1:

$\Delta L(\mu\text{m})$	FSR (nm)
70	8.19
170	3.37
470	1.22

IV. Conclusion

This work presents the design and simulation of several Mach–Zehnder interferometers (MZIs) based on silicon strip waveguides, with path-length differences ΔL of 70 μm , 170 μm , and 470 μm . The simulated transmission spectra demonstrate the expected interference behaviour and show that the free spectral range (FSR) decreases as the path-length difference increases. The simulation results presented in Figure 3 are in good agreement with the theoretical calculations, confirming the relationship between ΔL and FSR. These results validate the proposed MZI designs and provide a useful basis for selecting an appropriate path-length difference for future optical modulation and integrated photonic applications.

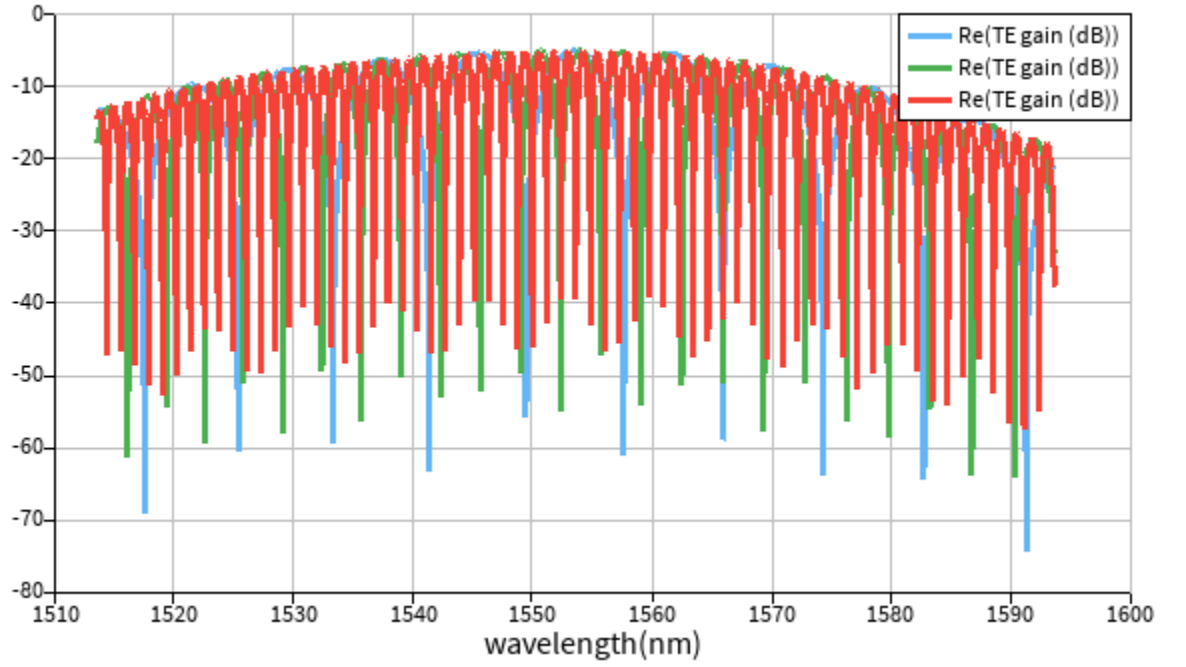


Figure 3: Spectrum of MZI with $\Delta L = 70$ (Blue), $\Delta L = 170$ (Green) and $\Delta L = 470$ (Red)